

DemoGen Genetics Laboratory

Hereditary Disease Panel — Laboratory Report

SYNTHETIC SAMPLE — NOT A REAL PATIENT

Patient: Sample Patient (synthetic) Sex: Female DOB: 1996-01-01
Specimen: Peripheral blood Report date: 2026-09-15 Report ID: DEMO-0001
Test: Hereditary disease panel (next-generation sequencing), 12 genes
Genes analyzed: BRCA1, BRCA2, CFTR, HBB, FBN1, LDLR, APOB, PCSK9, HFE, F5, F2, APOE

RESULT: POSITIVE — 2 clinically significant variants detected

Variant details

Variant 1
Gene: BRCA1 Variant: c.5266dupC (p.Gln1756Profs*74) rsID: rs80357906
Chromosome: 17 Zygosity: Heterozygous Classification: Pathogenic
Condition: Hereditary breast and ovarian cancer syndrome
Interpretation: Associated with increased inherited susceptibility to breast and ovarian cancer.

Variant 2
Gene: CFTR Variant: c.1521_1523delCTT (p.Phe508del) rsID: rs113993960
Chromosome: 7 Zygosity: Heterozygous Classification: Pathogenic
Condition: Cystic fibrosis (autosomal recessive)
Interpretation: Carrier of a pathogenic CFTR variant.

Additional genotypes

Gene	rsID	Genotype
APOE	rs429358	C/T
APOE	rs7412	C/C
TCF7L2	rs7903146	C/T
FTO	rs9939609	A/T
F5	rs6025	C/C

No pathogenic variants were detected in BRCA2, HBB, FBN1, LDLR, APOB, PCSK9, HFE or F2.

Methodology and limitations

Methodology: sequencing of coding regions and splice junctions of the genes listed above. Limitations: this test does not detect all types of genetic changes and does not assess chromosome number or structure. This document is a synthetic example created for an educational school project and does not describe any real person.